

📦 CARTON BANK #### Business Process Documentation & End-to-End Business Model **A Plastic Solution Initiative** *Circular Economy • Corrugated Carton Recovery • MSME Financial Inclusion* ---
Document Version: 1.0 **Prepared by:** Plastic Solution — Strategy & New Ventures
Geography: Federal Republic of Nigeria **Strategic Partners:** Pakam (Green-Tech) · LAPO MfB · Piggyvest · Paper Mills · EPR PROs



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1. Executive Summary

Carton Bank is a corrugated-carton (OCC) recovery and MSME-finance venture launched by **Plastic Solution** to convert recurring carton waste into **cash, savings, and working-capital loans** for Nigerian micro and small businesses.

The Opportunity

Nigeria generates an estimated **600,000 – 800,000 metric tonnes** of corrugated containers annually, with formal recovery below **35%**. The supply chain is dominated by informal, unbanked actors — creating a clear opening for a digitized, finance-enabled aggregator.

The Proposition

Carton Bank lets MSMEs:

#	Capability	Mechanism
1	Exchange carton waste for cash	Instant wallet credit at agreed grade-based price
2	Save earnings automatically	Co-branded "Carton SafeLock" with Piggyvest
3	Access working-capital loans	Disbursed by LAPO; repaid in carton deliveries
4	Track everything digitally	Pakam-powered app, USSD & WhatsApp bot
5	Earn ESG/EPR/Carbon recognition	Verified recovery sold to producer schemes

Headline Numbers (Year 1 → Year 3)

Metric	Year 1	Year 3
Tonnes recovered	8,000	36,000
MSMEs onboarded	1,500	12,000
Hubs operational	2	5
Monthly revenue (₦M)	~70	~520
EBITDA margin	Break-even	15%
CO ₂ e avoided (t)	26,000	120,000

2. Strategic Rationale

Macro Driver	Implication for Carton Bank
 Rising e-commerce (Jumia, Konga, Jiji, GIG Logistics)	Predictable, high-volume carton waste at known hubs
 FMCG & QSR boom (Chicken Republic, Domino's, ShopRite)	Daily OCC generation at fixed locations
 Paper-mill demand (BEL, Bel Impex, Dangote Paper)	Guaranteed offtake — Nigeria imports OCC today
 NESREA EPR Regulation (2022)	Compliance revenue + monetizable recovery credits
 Plastic Solution's existing logistics network	Shared fleet, hubs, agents → faster, cheaper launch
 Financial-inclusion gap among waste generators	Loan-against-waste is a defensible wedge

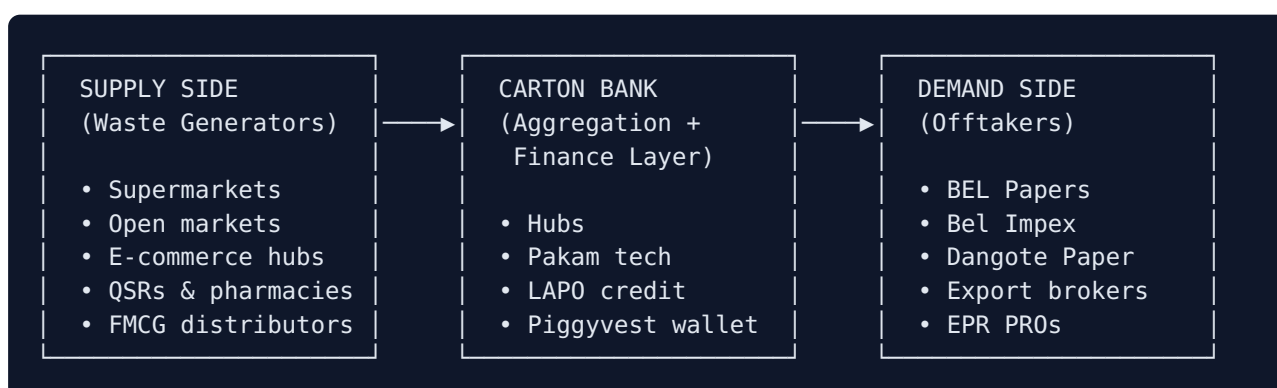
Competitive Edge: *Carton Bank is not "another collector." It is a **waste-as-collateral financial product** wrapped around a logistics operation.*

3. Business Model Canvas

3.1 Value Propositions

Stakeholder	What They Get
MSMEs	Cash, savings, working-capital loans from what they already throw away
Paper Mills	Reliable, baled, contamination-graded OCC at industrial scale
MFBs / Fintechs	A new, low-default borrower segment with carton-flow collateral
Government / EPR PROs	Auditable recovery tonnage for compliance
Pakam	Increased GMV and platform stickiness
Corporates / Brands	Scope 3 emissions data + verified EPR offset

3.2 Customer Segments



3.3 Revenue Streams

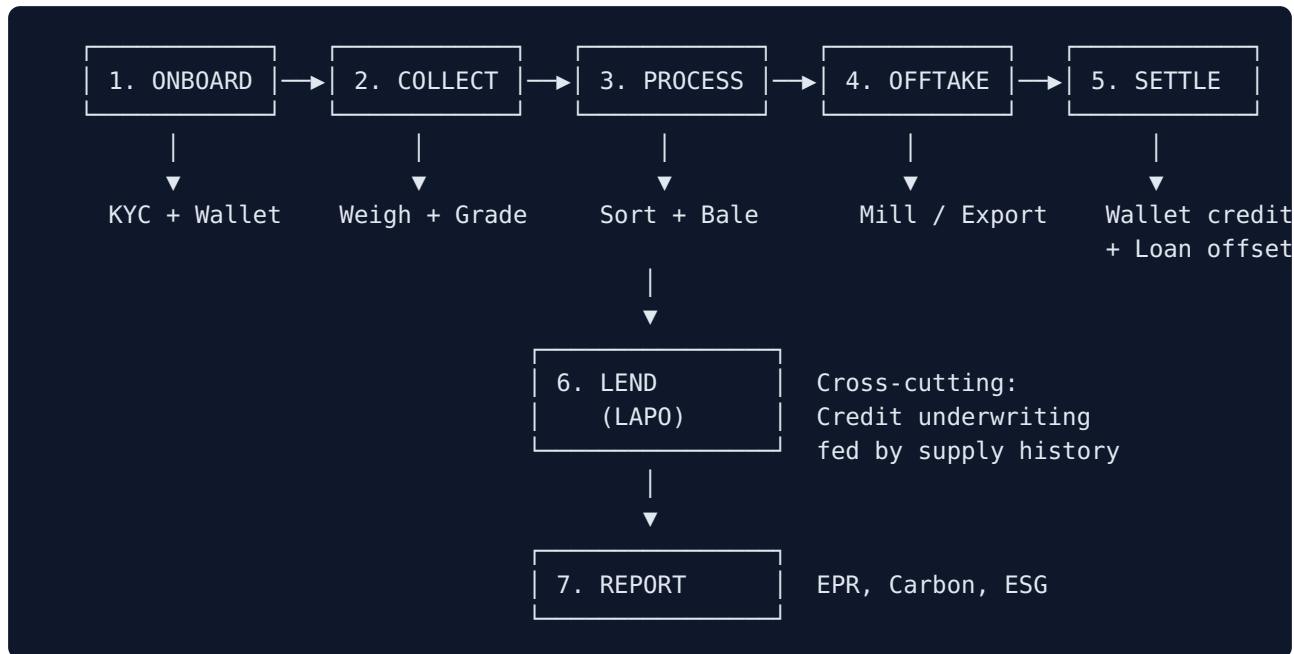
#	Stream	Margin Driver
1	OCC trading spread	Buy ₦60–₦90/kg → Sell ₦110–₦160/kg (baled)
2	Loan origination & interest share	2–3% origination + share of net interest
3	EPR credit sales	₦3,000–₦5,000 per verified tonne
4	Carbon credit monetization	~3.3 tCO ₂ e avoided per tonne OCC recycled
5	Service fees	Baling-as-a-service for large generators
6	Data & ESG subscriptions	Corporate dashboards for Scope 3 reporting

3.4 Cost Structure

Category	Type
OCC purchase price	Variable (largest line item)
Logistics (fuel, fleet, 3PL)	Semi-variable
Hub operations (rent, power, baling)	Fixed
Technology (Pakam fees, integrations)	Fixed
Loan-loss provisions	Variable
Compliance & licensing	Fixed
Marketing & agent commissions	Semi-variable

4. End-to-End Business Process

Process Overview



Stage 1 — Onboarding (KYC + Waste Profiling)

Channels: Field agents · Pakam app · USSD short code · WhatsApp bot · Hub walk-ins

Step	Action	Owner
1.1	Capture business name, BVN/NIN, geotagged address, premises photo	Field Agent
1.2	Estimate weekly carton output (2-week observation)	Field Agent + Generator
1.3	Tier the MSME — T1: <50 kg/wk · T2: 50–500 kg/wk · T3: >500 kg/wk	System
1.4	Issue Carton Bank ID + digital wallet linked to LAPO/Piggyvest	Pakam

Stage 2 — Collection

Three service tiers — generator chooses based on convenience vs. price:

Model	Price to Generator	Logistics Cost	Best For
 Drop-off at hub	Highest (Grade A premium)	None	Markets, large clusters
 Scheduled pickup	Mid	Routed	Supermarkets, restaurants
 On-demand pickup (app)	Lowest net	High	E-commerce, occasional users

Standard Operating Procedure:

1. Generator **flattens** cartons (price premium).
2. Agent **weighs** on calibrated digital scale.
3. **Contamination check** — reject wax-coated, food-soiled, laminated.
4. **Grade assigned** — A (clean OCC), B (mixed), C (rejected).
5. **QR-coded receipt** issued; wallet credited **in real time**.

Stage 3 — Aggregation & Processing

Step	Detail
3.1	Transport to nearest hub (start Lagos: Ikeja, Apapa, Lekki)
3.2	Sort by grade
3.3	Bale using horizontal/vertical balers (400–500 kg bales)
3.4	Store under cover — moisture <12% (mills reject wet bales)
3.5	Log to Pakam inventory: batch ID, weight, grade, source

Stage 4 — Offtake & Sales

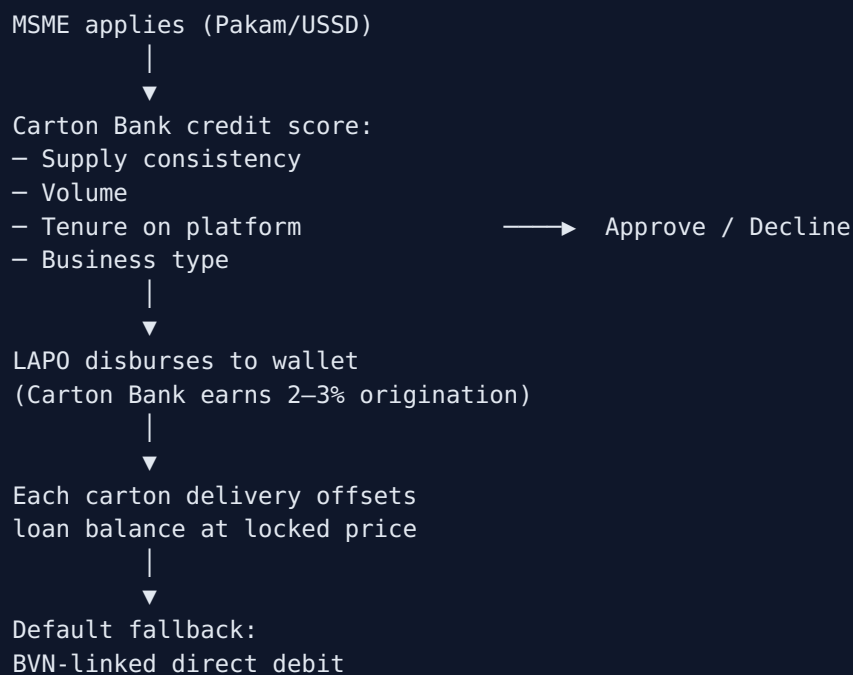
1. Weekly tonnage forecast issued to mills.
2. Contracts: **spot + 3-month fixed-price framework** with BEL Papers / Bel Impex.
3. Logistics to mill (own fleet or 3PL: Kobo360, Truckr).
4. Quality reconciliation at mill weighbridge.
5. Payment within **7–14 days** — invoice-discounting fallback via Sterling/Wema.

Stage 5 — Carton-Collateralized Lending ★ *The Differentiator*

Loan Product Specification

Parameter	Value
Ticket size	₦20,000 – ₦500,000
Tenor	30 / 60 / 90 days
Interest	4–6% flat per month (LAPO regulated)
Repayment	Weekly carton deliveries auto-deducted from wallet + optional cash top-up
Eligibility	≥8 weeks of verified carton-supply history

Lending Flow



Risk-share structure: Carton Bank guarantees a **10% first-loss tranche**; LAPO funds the rest. This is bankable and keeps incentives aligned.

Stage 6 — Monitoring, Reporting & EPR Monetization

Output	Frequency	Buyer / Recipient
Pakam ops dashboard	Real-time	Internal
Recovery report	Monthly	NESREA · LAWMA · EPR PRO
Carbon avoidance report	Quarterly	Carbon registry / buyers
ESG dashboard	Monthly	Corporate clients (Nestlé, NB, Friesland)

5. Key Partnerships

5.1 Pakam — Green-Tech Platform

- **Role:** Digital backbone — user/agent apps, wallet, geolocation, scheduling, ledger, ESG dashboard.
- **Commercial:** Revenue share (5–8% of transaction value) **or** SaaS + per-pickup fee.
- **Integration:** APIs to Carton Bank ERP, IoT weighbridges, MFB core banking.

5.2 LAPO Microfinance Bank

- **Role:** Licensed lender, KYC, disbursement, collections.
- **Commercial:** Carton Bank originates & services; LAPO funds. Net interest split **~60/40** (LAPO/Carton Bank). Carton Bank holds **10% first-loss**.

5.3 Piggyvest

- **Role:** Wallet + auto-savings ("Carton SafeLock").
- **Commercial:** Co-branded product, referral fee, float share.

5.4 Paper Mills (BEL • Bel Impex • Dangote Paper)

- **Role:** Primary offtakers.
- **Commercial:** Fixed-tonnage contracts with quarterly price review + volume rebates.

5.5 EPR PROs (FBRA • Alliance for Responsible Recycling)

- **Role:** Buy verified recovery tonnage for producer-obligation discharge.
- **Commercial:** ₦/tonne credit fee.

5.6 Logistics (Kobo360 • Truckr • In-house Fleet)

- **Role:** Inter-hub + hub-to-mill movement.
- **Commercial:** Per-trip or per-tonne-km rates.

6. Operational Footprint

Phase 1 Hubs (18 months):

#	Hub	Location	Throughput Target	Capex
1	HQ Hub	Ikeja, Lagos	400 t/mo	₦ 45M
2	Apapa Satellite	Apapa, Lagos	250 t/mo	₦ 25M
3	South-West Hub	Iwo Rd, Ibadan	200 t/mo	₦ 22M
4	South-East Hub	Ariaria, Aba	250 t/mo	₦ 25M
	Total		1,100 t/mo	₦117M

Standard Hub Spec: 600 sqm covered yard · 1 baler · 1 forklift · weighbridge · 4 field agents · 1 hub manager · 2 loaders.

7. Financial Plan & Obligations

7.1 Start-Up Capex (Year 0)

Item	₦M
Balers (4 horizontal units)	60
Forklifts (2)	18
Weighbridges + IoT scales	12
Trucks (3 × 10-tonne, used)	60
Warehouse setup & lease deposits	35
Pakam tech integration	15
Working capital (carton-buy float)	80
Loan-book seed (first-loss)	30
Branding, marketing, agent training	12
Licenses & registrations	8
Total	₦330M (~\$200–220k)

7.2 Monthly Opex (Steady State — Month 12)

Item	₦M
Carton purchases (1,100 t × ₦75k)	82.5
Salaries (40 staff incl. agents)	12.0
Logistics & fuel	9.0
Rent, utilities, repairs	5.0
Tech & Pakam fees	3.0
Loan-loss provisions	2.0
Marketing & community	2.0
Admin & compliance	2.0
Total	117.5

7.3 Monthly Revenue (Steady State — Month 12)

Stream	₦M
OCC sales (1,100 t × ₦135k)	148.5
Loan origination + interest share	4.0
EPR credits (1,100 t × ₦4k)	4.4
Carbon credits	1.0
Service / baling fees	1.5
Total	159.4

Gross margin: ~26% · **EBITDA margin (mature):** 12–15% · **Payback period:** 28–34 months

7.4 Funding Strategy

Source	Instrument	Amount
Plastic Solution	Internal equity	₦100M
DBN / BOI / AfDB	Green / impact debt	₦150M
GIZ · FCDO · CIO · Coca-Cola Foundation	Grants	₦50M
LAPO	Loan-book balance sheet (off-BS for us)	—
Sterling / Wema	Invoice-discounting working capital line	revolving

8. Key Considerations & Risk Register

8.1 Commercial Risks

Risk	Impact	Mitigation
OCC price volatility (global pulp index)	High	3-month forward contracts; price-collar formulas
Mill payment delays	Medium	Factoring facility; diversified buyers; export option
MSME loan defaults	Medium	Carton-flow collateral; LTV $\leq$ 60% of 90-day supply; MFB risk-share
Informal aggregator competition	Medium	Price-premium on Grade A; non-cash perks (loans, savings, insurance)

8.2 Operational Risks

Risk	Impact	Mitigation
Rainy-season moisture damage	High	Covered yards; moisture meters; dehumidified storage
Fire hazard (paper bales)	High	Insurance, suppression systems, NEMA-compliant siting
Weight fraud at collection	Medium	Calibrated tamper-seal scales; audit team; spot checks
Bale theft in transit	Medium	GPS-tracked trucks; QR-sealed bales

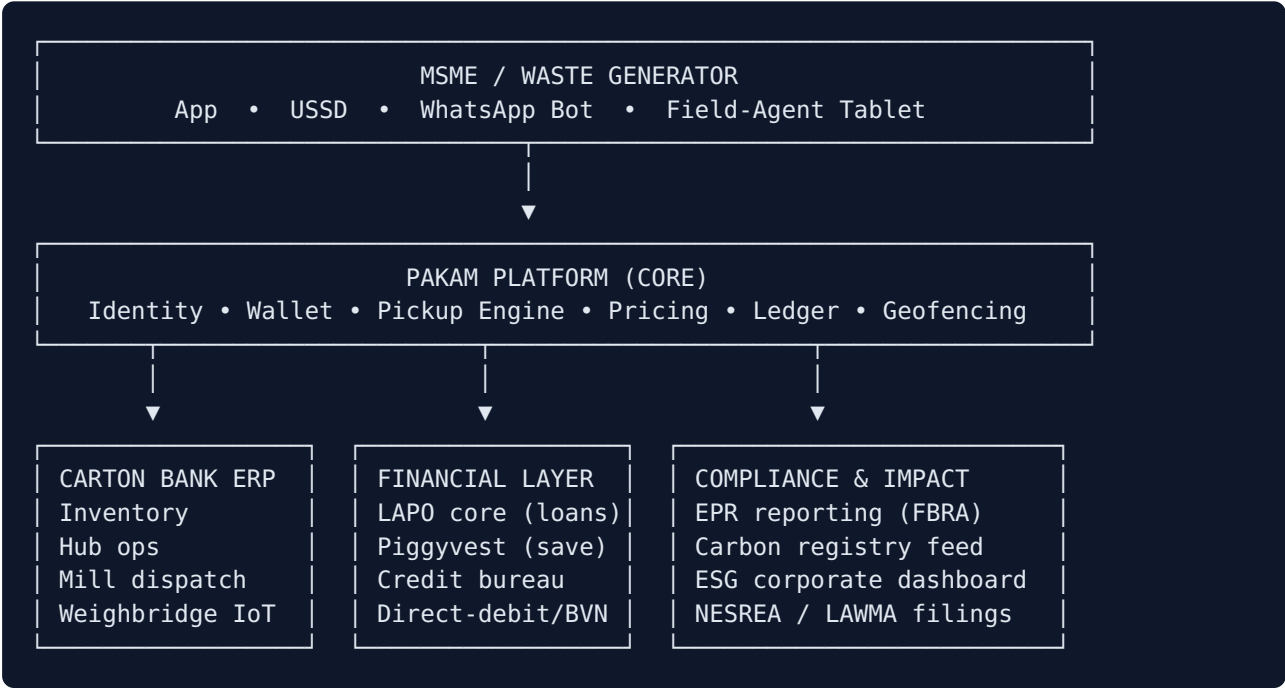
8.3 Regulatory & Compliance

-  Register with **NESREA** as a recycler
-  **LAWMA PSP** carton-recycler permit (Lagos)
-  **LASEPA** environmental impact compliance
-  Join an **EPR PRO** (e.g., FBRA) — mandatory under the NESREA EPR Regulation
-  **NDPR** data-privacy compliance (handled via Pakam)
-  **No direct lending license needed** — LAPO is the regulated lender. (*Critical structural point.*)

8.4 Strategic Risks

Risk	Mitigation
Cannibalization of Plastic Solution core focus	Separate P&L; dedicated GM; shared back-office only
Brand confusion	Sub-brand: <i>"Carton Bank — a Plastic Solution initiative"</i>
Donor/grant dependency	Operational profitability by Month 18; grants accelerate, not sustain

9. Technology Architecture

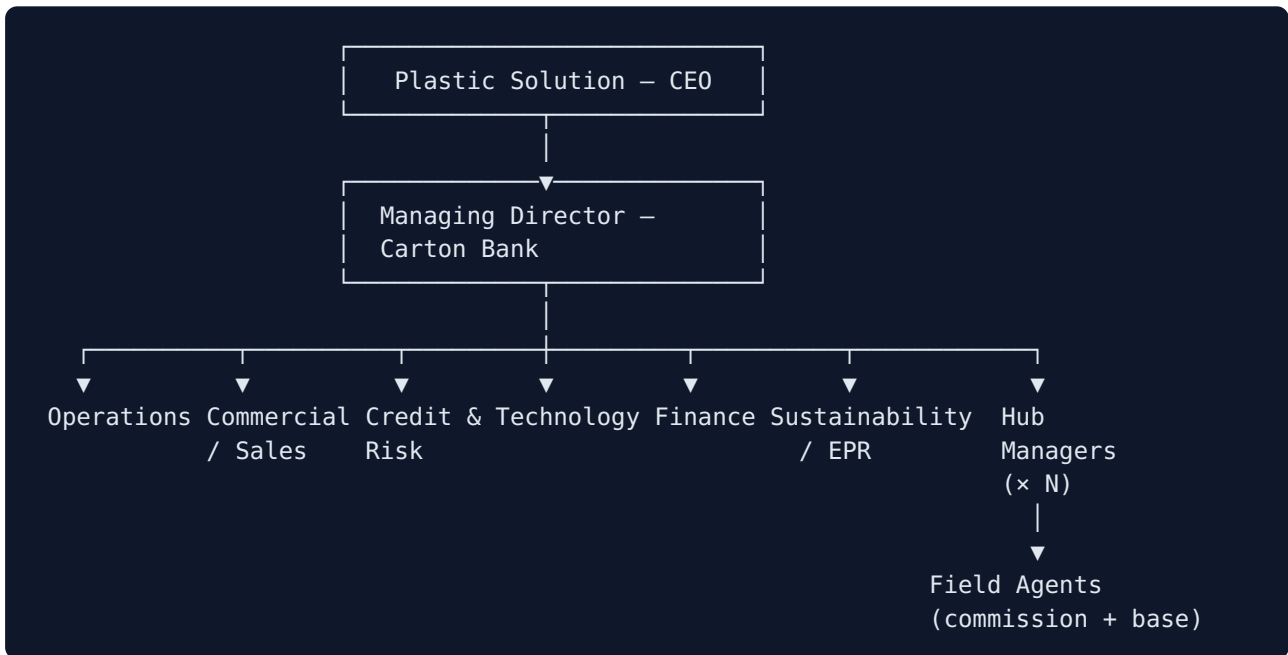


Key Capabilities: - Real-time weight & price capture - Wallet, savings, and loan ledger - Geofenced pickup routing - ESG / EPR dashboards - Waste-history-fed credit scoring engine

10. Go-to-Market Roadmap

Phase	Timeline	Milestones
0 — Foundation	M1–M3	Legal setup · MOUs (Pakam, LAPO, BEL) · Ikeja pilot hub · hire GM + 10 staff
1 — Pilot	M4–M6	200 MSMEs · 150 t/mo · 50 loans · prove unit economics
2 — Lagos Scale	M7–M12	1,500 MSMEs · Apapa + Lekki hubs · launch EPR sales · 500 active loans
3 — Regional Expansion	M13–M24	Ibadan + Aba hubs · 4,000 MSMEs · 1,100 t/mo · carbon-credit issuance
4 — National + Export	M25–M36	Kano, PH, Abuja hubs · OCC export pilot (Vietnam/India) · Series A raise

11. Organizational Structure



Shared with Plastic Solution: HR · Legal · Brand · Finance back-office.

12. KPIs & Governance

KPI	Year 1	Year 2	Year 3
Tonnes recovered	8,000	18,000	36,000
MSMEs onboarded	1,500	5,500	12,000
Active loans	500	2,000	5,000
Loan default rate (90+ DPD)	<8%	<6%	<5%
Gross margin	20%	24%	28%
EBITDA margin	Break-even	8%	15%
CO ₂ e avoided (t)	26,000	60,000	120,000
EPR tonnage sold	5,000	14,000	30,000

Cadence: Monthly ops dashboards · Quarterly board reviews · Annual ESG / impact report (co-published with Pakam).

13. Long-Term Vision

*Carton Bank evolves into "**Material Bank**" — a multi-stream circular-economy wallet covering OCC, PET, HDPE, aluminum, and e-waste — positioning **Plastic Solution as the dominant waste-as-collateral financial infrastructure in West Africa.***

Potential exits: - Strategic sale to an FMCG-backed Producer Responsibility Organization - Merger with a regional recycler - Listing on the NGX Growth Board

Why This Wins

Carton Bank's edge is **not** that it collects cartons — many do. Its edge is that it **turns recurring waste into a financial asset class** for MSMEs, underwritten by:

-  **Data** that Pakam captures
-  **Balance sheet** that LAPO / Piggyvest provide
-  **Logistics** that Plastic Solution already runs
-  **Compliance value** that NESREA EPR mandates

***Execution discipline** on pricing, moisture, loan underwriting, and mill relationships will determine whether this becomes a **N5B/year business by Year 5** — or a stalled pilot.*

Start lean** (one hub · one mill · one MFB). **Prove unit economics. Then scale aggressively.

14. Appendices

A. Glossary

Term	Meaning
OCC	Old Corrugated Containers — used corrugated cartons
EPR	Extended Producer Responsibility
PRO	Producer Responsibility Organization
MfB	Microfinance Bank
MSME	Micro, Small & Medium Enterprise
NESREA	National Environmental Standards & Regulations Enforcement Agency
LAWMA / LASEPA	Lagos State waste / environmental regulators
NDPR	Nigeria Data Protection Regulation
DPD	Days Past Due

B. Key External Contacts to Open

1. **Pakam** — Partnership & API team
2. **LAPO MfB** — Head of Digital Lending
3. **Piggyvest** — Business Partnerships
4. **BEL Papers / Bel Impex / Dangote Paper** — Procurement
5. **FBRA** — EPR programme office
6. **NESREA** — Recycler registration
7. **LAWMA** — PSP licensing
8. **DBN / BOI** — Green credit lines
9. **GIZ / FCDO Manufacturing Africa** — Grant programmes

C. Document Control

Version	Date	Author	Notes
1.0	2026-05-11	Strategy & New Ventures, Plastic Solution	Initial release

 Carton Bank *Where Waste Becomes Working Capital.* — End of Document —